7. Hidden Windows Repair Commands

CMD (Admin):

sfc /scannow

9. Installed Windows Date

CMD:

systeminfo | find "Original Install Date"

Overview of Peripherals

Theory Session

Learning Objectives

After this session, students will be able to:

- Understand what peripherals are.
- Identify different types of peripherals.
- Explain the functions of printers, scanners, and webcams.

- Understand common connection interfaces used to connect peripherals to a computer.

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1. Introduction to Peripherals

What are Peripherals?

Peripherals are devices that are connected to a computer to add extra functionality.

A computer can work without some peripherals, but these devices make tasks easier and more efficient.

Examples of Peripherals

Peripheral	Function
Printer	Prints documents and images on paper
Scanner	Converts paper documents into digital form
Webcam	Captures photos and videos
Keyboard	Used to enter data
Mouse	Used to control the pointer
Speakers	Output sound

Quick Activity

Ask students:

Can you name 5 devices connected to your computer?

Let students answer and classify them as peripherals.

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2. Printers

What is a Printer?

A printer is an output device that produces a hard copy of digital documents on paper.

Uses of Printers

- Printing assignments
- Printing photographs
- Printing bills and invoices
- Printing certificates

Types of Printers

1. Inkjet Printer

- Uses liquid ink.
- Good for color printing.
- Common in homes and schools.

Advantages:

- Low cost
- Good image quality

2. Laser Printer

- Uses toner powder.
- Faster than inkjet printers.
- Suitable for offices.

Advantages:

- Fast printing
- Lower cost per page

Interesting Question

Which printer is better for printing 500 pages daily?

Answer:
Laser Printer

Because it is faster and more economical for large printing jobs.

---

3. Scanners

What is a Scanner?

A scanner is an input device that converts physical documents into digital files.

Uses of Scanners

- Scanning certificates
- Scanning photographs
- Creating PDF documents
- Digitizing old records

Working of a Scanner

1. Place document on scanner glass.
2. Scanner captures the image.
3. Software converts it into a digital file.
4. File is saved on the computer.

Types of Scanners

Flatbed Scanner

- Most common scanner.
- Document is placed on glass.

Handheld Scanner

- Moved manually over the document.

Document Scanner

- Used for scanning multiple pages quickly.

Real-Life Example

When you upload an Aadhaar card or PAN card online, it is often scanned using a scanner.

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4. Webcams

What is a Webcam?

A webcam is an input device used to capture photos and videos.

Uses of Webcams

- Online classes
- Video conferencing
- Live streaming
- Security monitoring

Examples

- Zoom meetings
- Google Meet classes
- Microsoft Teams meetings

Features of a Webcam

- Resolution (720p, 1080p, 4K)
- Built-in microphone
- Auto focus

Fun Question

Why do laptops usually have webcams built in?

Answer:
To allow users to join video calls without connecting an external camera.

---

5. Connection Interfaces

What is a Connection Interface?

A connection interface is the method used to connect a peripheral device to a computer.

Common Interfaces

1. USB (Universal Serial Bus)

Most common connection type.

Examples:

- Printer
- Webcam
- Scanner
- Pen Drive

Advantages:

- Easy to connect
- Plug and Play

2. Bluetooth

Wireless connection technology.

Examples:

- Wireless mouse
- Wireless keyboard
- Headphones

Advantages:

- No cables required

3. Wi-Fi

Allows devices to connect through a network.

Examples:

- Wireless printers
- Network scanners

Advantages:

- Access from multiple computers

4. HDMI

Used for video and audio transmission.

Examples:

- Monitor
- Projector

Advantages:

- High-quality video and sound

Quick Comparison

Interface	Wired/Wireless	Example Device
USB	Wired	Printer
Bluetooth	Wireless	Mouse
Wi-Fi	Wireless	Network Printer
HDMI	Wired	Projector

Activity 1: Identify the Device

Show images or actual devices and ask:

1. Which device prints documents?
2. Which device converts paper to digital format?
3. Which device is used for video calls?
4. Which connection is wireless?
5. Which interface is commonly used for printers?

Answers:

1. Printer
2. Scanner
3. Webcam
4. Bluetooth/Wi-Fi
5. USB

Revision Questions

1. What is a peripheral device?
2. Is a printer an input or output device?
3. What is the function of a scanner?
4. Why is a webcam used?
5. Name two types of printers.
6. Which printer is faster: Inkjet or Laser?
7. What is the most common interface used for peripherals?
8. Give one example of a Bluetooth device.
9. What is the use of HDMI?
10. Name any three peripherals.

`start microsoft.windows.camera`: Open Camera Quickly
systeminfo -Find Computer Hardware Information

Interesting Peripheral Practicals

These activities are simple, fun, and help students understand peripherals by actually using them.

Practical 1: Check Connected USB Devices

Command

```
wmic path Win32_USBControllerDevice get Dependent
```

What Students Learn

- Which USB devices are connected to the computer.
- Examples: Mouse, Keyboard, Pen Drive.

Can you identify your mouse or keyboard in the list?

Practical 2: View Installed Printers

Command

```
wmic printer get name,status
```

Example Output

```
Name                        Status
HP LaserJet                 OK
Microsoft Print to PDF      OK
```

What Students Learn

- Which printers are installed on the computer.
- Difference between a real printer and a virtual printer.

Practical 3: Open the Webcam

Command

```
start microsoft.windows.camera:
```

Activity

- Open the Camera app.
- Take a photo.
- Record a short video.

What Students Learn

- How a webcam works.
- How webcams are used in online meetings and classes.

Practical 4: View System Information

Command

```
systeminfo
```

Activity

Find:

- Computer Name
- Operating System
- RAM

What Students Learn

- Basic information about the computer.

Practical 5: Check Storage Devices

Commands

```
diskpart
```

Then:

```
list volume
```

To exit:

```
exit
```

What Students Learn

- View hard drives and pen drives connected to the system.
- Understand storage devices as peripherals.

Practical 6: Open Device Manager

Command

```
devmgmt.msc
```

Activity

Find:

- Keyboard
- Mouse
- Webcam

- Printers

What Students Learn

- How Windows manages hardware devices.

Practical 7: View Installed Printers

Command

```
control printers
```

Activity

- View all installed printers.
- Find the default printer.

Ask Students

Which printer is set as the default printer?

Practical 8: Check Audio Devices

Command

```
mmsys.cpl
```

Activity

- Check speakers.
- Check microphones.

What Students Learn

- Speakers and microphones are also peripherals.

Practical 9: Print a File Using Command

Step 1

Open Notepad:

```
notepad
```

Step 2

Type:

```
Hello Students
Today we are learning about Peripherals.
```

Step 3

Save as:

```
test.txt
```

Step 4

Print the file:

```
notepad /p test.txt
```

What Students Learn

- How a printer can be used from the command line.

Practical 10: View Installed Drivers

Command

```
driverquery
```

What Students Learn

- Every hardware device needs a driver.
- Drivers help Windows communicate with peripherals.

Ask Students

What will happen if a printer driver is not installed?

Answer:
The printer may not work properly.

Quiz

Device	Input or Output?
Keyboard	Input
Mouse	Input
Scanner	Input
Webcam	Input
Printer	Output
Speaker	Output
Microphone	Input
Touch Screen	Both

Open these tools using commands only:

Printers

```
control printers
```

Device Manager

```
devmgmt.msc
```

Sound Settings

```
mmsys.cpl
```

Camera

```
start microsoft.windows.camera:
```